

## Purpose

To build a secure, scalable system that allows students to apply, register, enroll in courses, pay tuition, access academic records, and interact with faculty—fully online.

The platform will support administrative workflows, compliance requirements, faculty management, and degree audit functionalities.

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## Role : (System User)

- Admin ( BOT + VC + Pro VC )
- Academic Register ( Register + Dean + Department Head )
- Course Co-ordinator
- Faculty
- Student

## Scope :

The system will support:

- Full student lifecycle: inquiry → application → admission → enrollment → graduation
- Fully online degree programs
- Faculty onboarding, course creation, grading workflow
- Secure document storage and verification
- Compliance with regulatory bodies (e.g., THECB, SACSCOC requirements)
- Integration with payment gateways, LMS, video conferencing tools

# Functional Requirement

## Student :

### Application Portal

- Apply online ,
- Dashboard :
  - Register | Sign in to their profile
  - Add documents : ( Certificate , Documents , Image )
  - Profile dashboard after registration
- Automated eligibility check
- Pay for apply ( Stripe , Paypal etc )
- Application tracking ( Submitted , under review , grant )
- Automated Email notification if got selected
- Automated acceptance letter sent or email sended if got selected .

### Student Portal ( once selected student will got a Unique ID )

- **Student can work on their profile :**
  - Image
  - Name
  - Address
  - Parents address | contact | identity
  - Degree ( Which degree ) .
  - Enrolled Semester
- **Course work ;**
  - Select eligible subject - class - Time before the semester started , Once semester started , it cannot be changed
  - Can drop requests to drop the course .
  - Check for attendants .
  - Check Marks - Marks history - Petition - recheck request .
  - Check grade history - Grade revisit
  - Check degree history
  - Apply - Submit for papers and documents

- ~~— Seat for exam — submit / upload assignment~~
- Apply for certificate
- Contact teacher - faculty - register
- Take appointment
- Ask for Leave

## Faculty portal :

Faculty will also have a similar profile portal

- Image
- Name
- Address
- Parents address | contact | identity
- Degree ( Which degree ) .
- Enrolled Semester

## Course portal

- Created by register
- Can check - update students
- Request for Student adding - dropping
- Check the current enrolled student's profile .
- ~~— Upload course documents — course materials — course topics~~
- ~~— Create assignment — exam~~
- Announce topics
- ~~— Check added assignments — grade~~
- Submit marks - grade
- Ask for marks - grade edit with proof.
- Track student attendants .

## Faculty Portal ( Google classroom - Canvas )

- Dashboard of assigned courses
- Upload syllabus, materials
- Grade book & rubrics
- Publish grades (midterm & final)
- Track attendance

- Message students & announcements
- Course evaluation reviews (student feedback)

## Program & Course Management

### Admin Register can:

- Create degree programs (credits, duration, curriculum rules)
  - Create courses (code, title, faculty, credit hours, syllabus)
  - Set prerequisites & corequisites
  - Assign faculty to courses
  - Manage academic calendar (semesters, start/end dates)
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## Registration & Enrollment System

### Students must be able to:

- View available programs and courses
- Enroll in courses per term
- Check seat availability
- See prerequisites and warnings
- Request add/drop
- Automatic conflict checking (time/credits/prerequisite rules)
- Tuition & fee calculation based on enrolled credits
- Waitlist system

## Academic Register :

- Approve/deny special enrollment requests
- Enforce full-time/part-time load rules
- Manage course capacities
- Change Grade ( Based on the teacher request with proof )
- Create degree programs (credits, duration, curriculum rules)
- Create courses (code, title, faculty, credit hours, syllabus)
- Set prerequisites & corequisites
- Assign faculty to courses
- Manage academic calendar (semesters, start/end dates)

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## Finance & Payment

- Tuition calculation logic per credit/program
- Scholarships & financial aid integration - Based on the quota and grade
- Invoices & payment plan options
- Online payments via Stripe/PayPal/ACH
- Auto reminders for overdue fees
- Refund rules management
- Download receipts

## Compliance Features

- FERPA compliance
- Data encryption (at rest & in transit)
- Document verification & authenticity
- Secure transcript management
- Audit trails for every change
- Identity verification (KYC, proctoring)

## LMS Integrations

Integration with:

- Moodle / Canvas / Blackboard / **Google Classroom**
- Zoom / **Google Meet** for live classes
- Proctoring tools (ProctorU, Examity)
- Plagiarism detection (Turnitin)

APIs must be designed for plug-and-play with different LMS providers.

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## Communication System

- Announcements (global or per course)
- Email/SMS notifications
- In-app messaging
- Discussion forums
- Push notifications (mobile apps)

## Reporting & Analytics - Admin dashboard

- Student performance dashboards
- Enrollment trends
- Graduation rates
- Faculty performance metrics
- Financial reports
- Regulatory reports (credit hours, student FTE, course completion)

## Non-Functional Requirements

### Security

- JWT-based authentication
  - Role-based access control
  - Multi-factor authentication
  - Data encryption
  - Daily backups - Physical backup server ( multiple layer )
  - Penetration testing
  - GDPR/FERPA compliance
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## Performance & Scalability

- Handle 50,000+ concurrent users
  - Auto-scaling cloud environment
  - API response time < 500ms
  - Database optimized for large student records
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## Usability

- Mobile-first responsive design
  - Accessible (WCAG AA compliant)
  - Simple onboarding UX
  - Multi-language support (English + option to expand)
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## Reliability

- 99.9% uptime
  - Fault-tolerant architecture
  - Disaster recovery plan
  - Load-balanced servers
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## System Architecture Requirements

- Microservices-based architecture : lets start with Monolithic .
- React/Next.js front-end
- Node.js/Python back-end
- PostgreSQL or MySQL database
- Redis caching
- AWS/GCP/Azure deployment

- REST + GraphQL APIs
  - ~~Optional blockchain-based credential verification~~
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